

29 - Intro to A Level organic chemistry

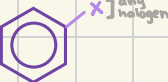
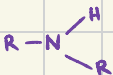
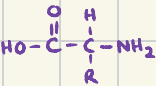
aliphatic compounds

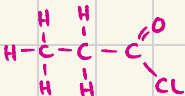
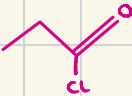
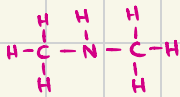
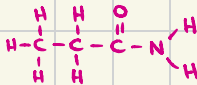
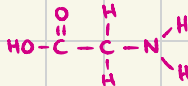
↳ organic compounds w/o a benzene ring
can be straight or branched chains, or non-aromatic rings

aromatic compounds

↳ organic compounds with a benzene ring

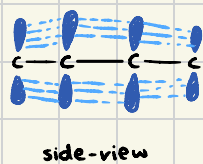
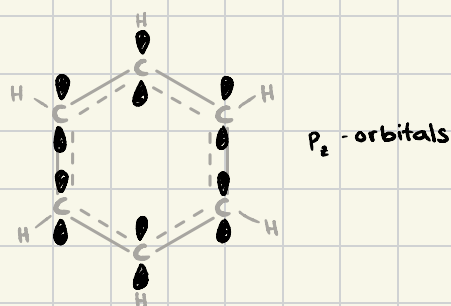
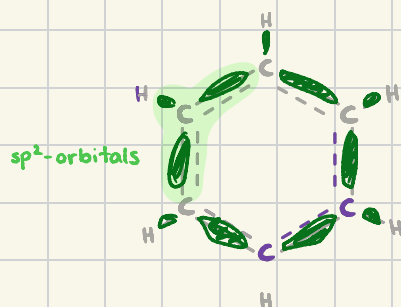
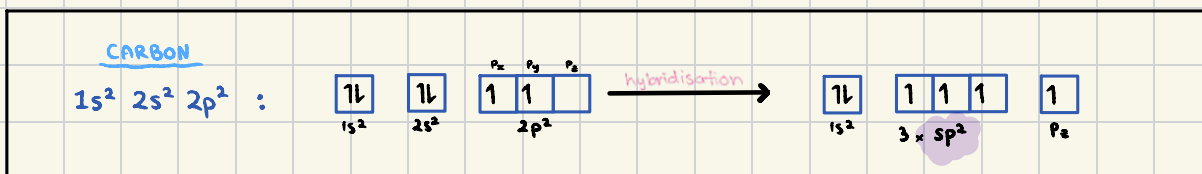
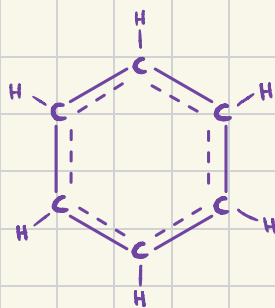
Functional group is a group of atoms in an organic compound that determine the chemical & physical properties

Homologous series	Functional group	Structural formula
arene	arene	
halogenoarene	halogen	
phenol	phenol	
acyl chloride	acyl chloride	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{Cl}$ rest of the compound
amines	amine	
amides	amide	$\text{R}-\overset{\text{O}}{\parallel}{\text{C}}-\text{NH}_2$
amino acids	amine & carbox. acid	

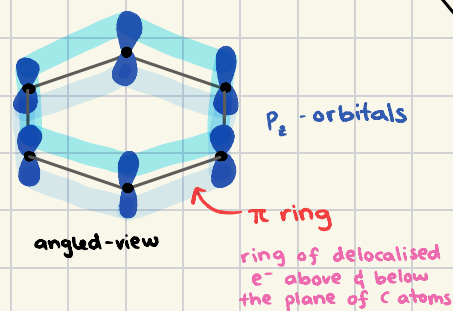
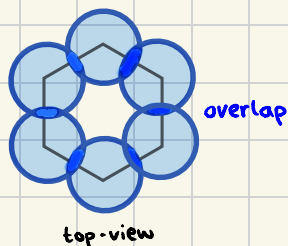
e.g	name	displayed formula	skeletal formula
	benzene	N/A	
	chlorobenzene	N/A	
	phenol	N/A	
	propanoyl chloride		
	Naming not required +		
	propanamide		
	2-aminoethanoic acid		

Benzene (C_6H_6)

↳ planar, cyclic molecule with a hexagonal shape



sideways overlapping
of the p_z -orbitals
↑
 π bond



★ π ring is very stable

★ it does not undergo addition (cause that would involve breaking the delocalisation & losing stability)

★ it will undergo substitution reactions (to preserve π ring)

Optical isomerism

also known as enantiomers

molecules that are non-superimposable images of each other

similarities b/w a pair of enantiomers

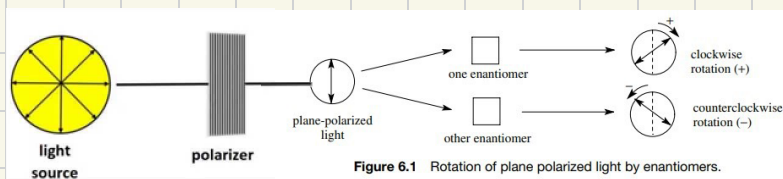
- ★ identical physical properties
- ★ identical chemical properties

differences b/w a pair of enantiomers

- ★ rotation of plane-polarised light
- ★ biological activities

Optically active : a substance that can rotate the plane of polarised light.

rotation of plane-polarised light



biological activities

→ enzymes/receptors

due to the chirality of biological systems, enantiomers may interact differently with these systems, leading to different biological effects

e.g one enantiomer of a drug may be active, while the other is inactive or even harmful

Racemic mixture : a mixture containing equal amounts of both enantiomers of a chiral compound

- ↳ the rotation of both enantiomers cancel each other out
- ↳ the racemic mixture does NOT rotate plane-polarized light
- ↳ it is **OPTICALLY INACTIVE**

Relevance of chirality in drug synthesis :

1) different biological activity of enantiomers

- many enzymes/receptors are chiral, which means they can differentiate b/w enantiomers
- one enantiomer of a drug might bind effectively to a receptor & produce a therapeutic effect
- the other enantiomer might be inactive or produce undesirable effects

e.g. Thalidomide → one enantiomer has therapeutic effects while the other causes birth defects

2) separation of a racemic mixture

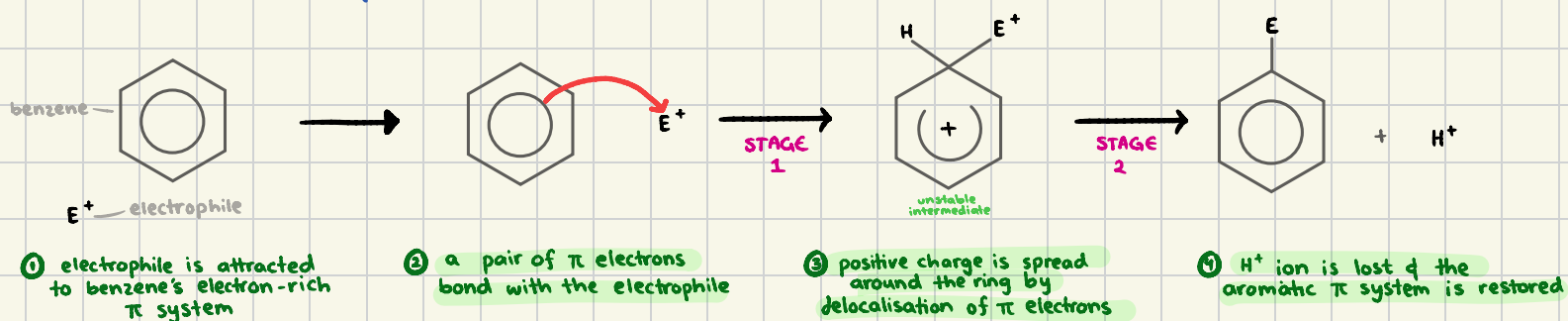
- since enantiomers can have different biological effects, it's necessary to separate a racemic mixture into its pure enantiomers
- chiral resolution : separating enantiomers through their difference in interactions with other chiral substances
- enantiomer-specific synthesis : developing synthetic routes that directly only produce the desired enantiomer

3) use of chiral catalysts

- to avoid the need to separate enantiomers, chemists can use chiral catalysts
- these catalysts are chiral themselves & can produce one enantiomer preferentially
- which leads to the formation of a single pure optical isomer
- crucial in drug manufacturing, where producing only the active enantiomer is efficient & safe

Types of organic mechanisms :

1) electrophilic substitution



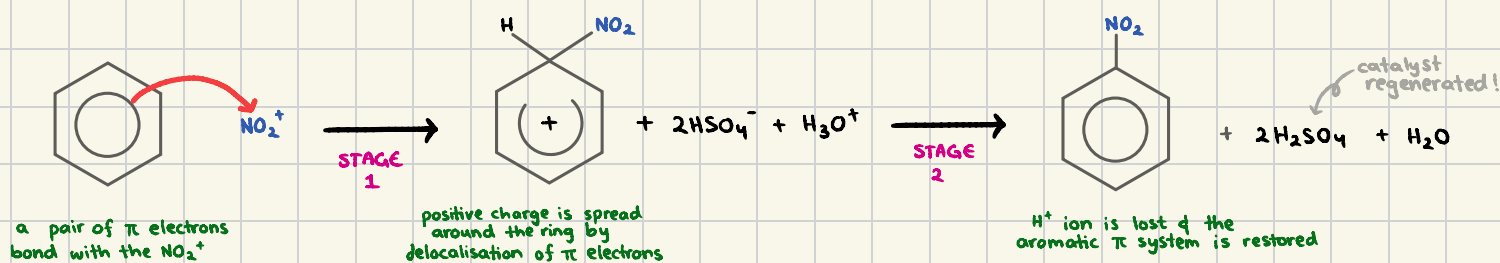
e.g Nitration of benzene

1) electrophile (NO_2^+) is generated by creating a mixture of conc. H_2SO_4 & conc. HNO_3

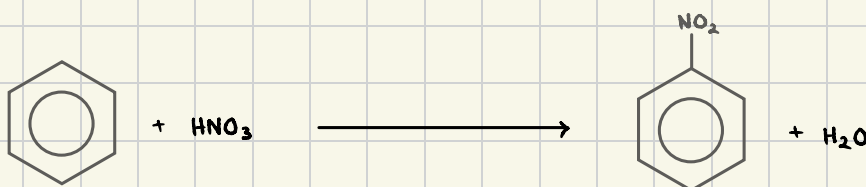


2) the nitrating mixture is mixed with benzene

mechanism :

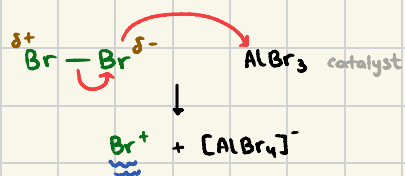


overall reaction :

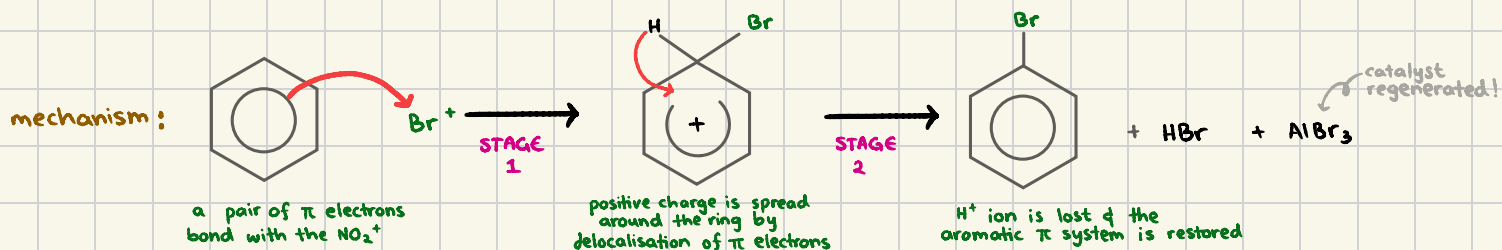


e.g Bromination of benzene

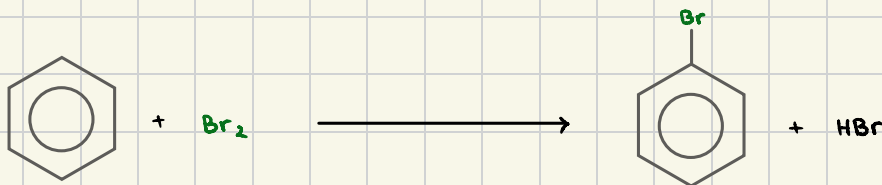
1) electrophile (Br^+) is generated by a polarising catalyst (e.g AlBr_3)



2) electrophile attacks the delocalised π bond system



overall reaction:



2) addition-elimination

